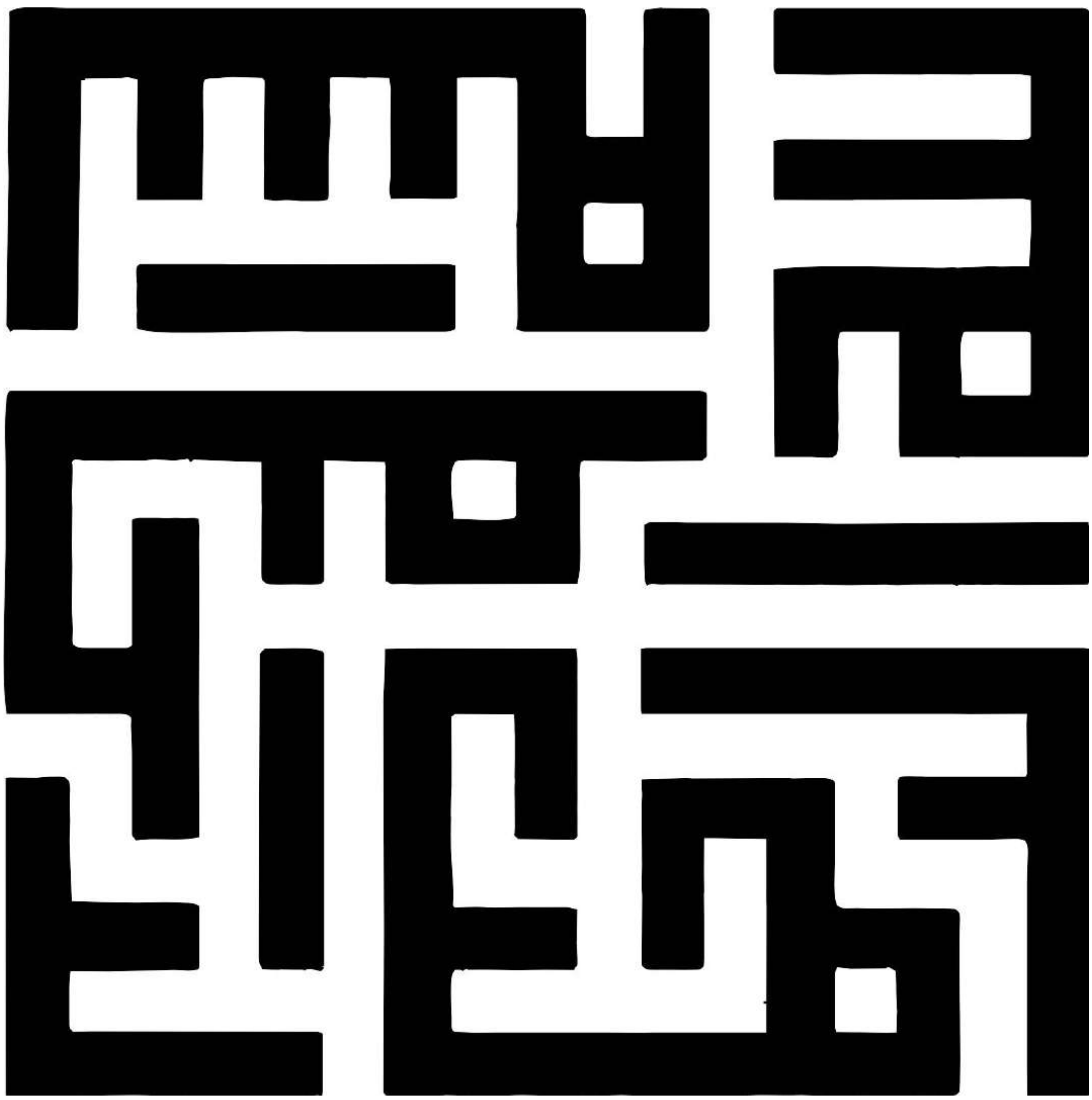

Impact of Vertex Degree on Exploration-Imitation Dynamics in Complex Systems

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Abstract
Motivated by Social Learning Theory and prestige-biased transmission (where high-status individuals attract greater attention and exert stronger influence on potential imitators) we study exploration–imitation dynamics of continuous strategies on networks, using vertex degree as a minimal proxy for both visibility (being selected as a role model) and imitation strength (effective influence). Building on the analytical baseline, we perform agent-based simulations on several canonical network models. In the baseline model, each agent selects a neighbor uniformly at random and updates its strategy as a combination of its previous state, the chosen neighbor's state with constant imitation strength α, and a stochastic exploration term. Exploration represents exogenous information and intrinsic behavioral fluctuations, which broaden the strategy distribution, while soft imitation counteracts this tendency by reducing dispersion; the competition between these mechanisms leads to a non-trivial steady-state variance. In the extended models, degree effects are introduced in two distinct ways: (i) by biasing role-model selection toward high-degree neighbors, and (ii) by making imitation strength explicitly degree-dependent. A fourth model incorporates both mechanisms simultaneously. To obtain statistically reliable estimates, each network realization is simulated multiple times and ensemble averages together with standard errors are reported; moreover, to disentangle dynamical effects from structural randomness, the same generated network realization is used across all four models. Our results show that, except for the well-mixed case, the considered networks are approximately matched in average degree, yet they produce markedly different stationary dispersions: networks with higher clustering and especially larger average shortest-path length exhibit larger steady-state variance. For Watts–Strogatz networks, the variance displays an initial transient dip before converging to its stationary value, and the dip depth is found to be inversely related to the algebraic connectivity of the network. Finally, for Barabási–Albert networks and for a range of the construction parameter m, the relative steady-state variance gap between the degree-dependent imitation model and the baseline model follows an exponential decay as m increases.

Keywords
social learning theory, prestige bias, exploration–imitation dynamics, agent-based modeling, complex networks, clustering, average path length, algebraic connectivity

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